

Chapter	Title	Total periods	Duration
Ch 8	Journey Inside the Atom	6	50 mins × 6

C No.	Targeted NCF competencies	Justification
C-1.1	Describes classification of elements in the Periodic Table, and explains how compounds (including carbon compounds) are formed based on atomic structure (Bohr's model) and properties (valency)	The chapter's whole domain is atomic structure via the Bohr model and the combining capacity (valency) that follows from electronic configuration — the 'atomic structure (Bohr's model) and properties (valency)' clause of C-1.1. The periodic-table / compound-formation clause of C-1.1 is not developed here; it is the domain of the sibling chapter (Atomic Foundation of Matter). The chapter's four sub-topics (model evolution, atomic/mass number and electronic configuration, valency, isotopes/isobars) are all facets of atomic structure and serve this single competency; no distinct second strand exists within CG-1 (mixtures C-1.2 and chemical equations C-1.3 are absent).

This assessment is formative. Every question traces to a specific learning outcome, and each outcome is derived from an NCF competency this chapter develops. Rather than testing recall or asking whether a student “understood” a topic, each item asks for an observable demonstration — a response, a classification, a justification, or a product — of the thinking the lessons built. The questions are grouped by sections, and each shows the learning outcome it assesses and its cognitive demand, so you can see exactly what each question checks and why.

SECTION 1

8.1 Rediscovering the Roots of Atomic Theory

Period 1	50 min	From Parmanu to Dalton: Tracing the Origins of Atomic Thinking
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Materials: Textbook, Blackboard or whiteboard

Teacher notes: This is the chapter's entry point — students arrive with everyday intuitions about matter being continuous, so naming the philosophical roots before experimental science helps them see why the question was hard. A common confusion to watch for: students may conflate Dalton's scientific claim with the Greek philosophical claim; press them to articulate the difference between a reasoned argument and a testable proposal. A useful checkpoint is whether students can correctly place each thinker on the experiment/reasoning axis before moving on.

8 min	The teacher draws a three-column timeline on the board (ancient India, ancient Greece, 1808) and introduces the question: how did people first explain what matter is made of? Students suggest what 'indivisible' might mean before any formal definition is given.
14 min	The teacher traces Kanada's concept of parmanus combining into dyads and triads, then Leucippus and Democritus's atomos — both arrived at independently, both philosophical rather than experimental. Students note the parallel and discuss: what does it mean that two unconnected traditions reached the same idea?
16 min	Dalton's 1808 scientific treatment is introduced as the turn from philosophy to testable proposal. Students compare the three accounts — Kanada, Greek atomists, Dalton — on a two-axis chart: (1) based on experiment or reasoning? (2) atoms combine or not? They justify each cell.
12 min	Pairs share their charts and the class reconciles differences. The teacher closes by surfacing what Dalton's proposal still left unanswered: if atoms are indivisible, what are they made of? Students write one question they predict the chapter will need to answer.

Homework: Find one modern scientific claim and one philosophical claim about the natural world; write two sentences explaining how you would distinguish between them.

Assessment

Q1 · Learning Outcome

Students can classify claims about natural phenomena as philosophical reasoning or experimentally grounded proposals and justify the distinction.

Cognitive Demand
Understanding

Kanada's concept of parmanu and the Greek idea of atomos both proposed that matter is made of indivisible particles. Which statement best describes what distinguishes Dalton's 1808 proposal from these earlier ideas?

- A. Dalton grounded his atomic proposal in experimental and chemical evidence rather than philosophical reasoning alone. ✓
- B. Dalton proposed that atoms combine into dyads and triads, whereas earlier thinkers claimed atoms never combine.
- C. Dalton was the first thinker to suggest that matter is made of indivisible particles.
- D. Dalton's atoms carry electrical charge, which the Greek atomos did not.

Answer section

Correct answer: A

What each choice reveals

- C Confuses originality of the indivisibility idea with Dalton's actual contribution; ignores Kanada and the Greek tradition.
- B Attributes the dyad/triad combining model specifically to Dalton, when it was Kanada's description; shows confusion between the two traditions.
- D Imports a property from later atomic discoveries (Thomson's electron work) into Dalton's original model, where charge played no role.

SECTION 2

8.2 A Short Historical Journey Through Atomic Models / 8.2.2 Testing Thomson's model: The gold foil experiment

Period 2	50 min	Electrons, Plum Pudding, and the Gold Foil: Two Models Compared
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Materials: Textbook (Fig. 8.4 gold foil diagram), Rulers, Plain paper for student diagrams

Teacher notes: Having established that Dalton saw atoms as indivisible, this unit moves into the experimental evidence that broke that picture open — connecting to the unanswered question students raised about atomic composition. The most persistent misconception here is that Rutherford's model simply 'improved' Thomson's rather than falsified a specific prediction; press students to name the exact observation — sharp backscattering — that ruled out the plum-pudding picture entirely. A good checkpoint is whether students can independently state all three Rutherford conclusions before the class closes.

10 min	The teacher briefly presents Thomson's discovery of the electron via cathode ray tubes and introduces the plum-pudding model. Students sketch the model from the description — uniform sphere of positive charge, electrons distributed throughout — and predict: if you fired a positive particle at this kind of atom, what path would it take?
15 min	The teacher reveals the gold foil experimental setup from the textbook figure. Students examine it and, before learning the results, draw a labelled prediction diagram showing expected deflection angles if Thomson's model were correct. Groups share their diagrams and agree on a class prediction.
13 min	The teacher describes the three actual observations: most particles pass through undeflected, some deflect at large angles, a few bounce straight back. Students compare their prediction diagrams against the real results and identify specifically which observation Thomson's model cannot explain and why.
12 min	Students construct Rutherford's three conclusions from the evidence in their own words: empty space, concentrated nucleus, orbiting electrons. They then evaluate: which observation maps to which conclusion? Each group writes one sentence per conclusion linking evidence to inference, and pairs cross-check.

Assessment

Q2 · Learning Outcome

Students can evaluate a scientific model by identifying the specific experimental observation that falsifies it and deriving a revised model from the evidence.

Cognitive Demand

Analysis

In the gold foil experiment, Geiger and Marsden observed that most alpha particles passed through the gold foil undeflected, some were deflected at large angles, and a very small number bounced almost straight back. Thomson's model predicted only slight deflections.

- (a) Identify the single observation from the gold foil experiment that is sufficient to rule out Thomson's plum-pudding model, and explain precisely why that observation is incompatible with a uniformly distributed positive charge.
- (b) State Rutherford's three conclusions from the experiment and link each conclusion to the specific observation that supports it.

Answer section**Look for**

- Identifies sharp backscattering (very small number of alpha particles bouncing back) as the falsifying observation, not merely 'large deflections'.
- Explains that a diffuse positive charge cannot exert the concentrated repulsive force needed to reverse a fast, massive alpha particle's direction.
- States all three Rutherford conclusions: most of the atom is empty space; positive charge and most mass are concentrated in a small nucleus; electrons orbit the nucleus.
- Correctly links: undeflected passage → empty space; large-angle deflection and backscattering → concentrated nucleus; orbital motion → electron positions.
- Avoids conflating the gold foil result with Thomson's prediction — the response distinguishes what was predicted from what was observed.

SECTION 3**8.2.3 Bohr's model of the atom**

Period 3

50 min

Bohr's Shells and the Stability Problem
Rutherford Could Not Solve

Materials: Textbook, Concentric-circle shell diagrams (pre-drawn or student-drawn), Blackboard

Teacher notes: Having built Rutherford's model through the gold foil evidence, this unit addresses the collapse problem that model introduced — a direct content connection. Students frequently misread Bohr's stationary states as meaning electrons are stationary; clarify that the electron still moves in its orbit but does not radiate energy while in that orbit. A useful probe is to ask students what would need to be true about the energy of an emitted photon if an electron falls from the M shell to the L shell — this tests whether they understand the quantised energy difference, not just the shell labels.

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| 10 min | The teacher restates Rutherford's stability problem — classical physics predicts an orbiting electron must radiate energy and spiral into the nucleus — without yet giving Bohr's solution. Students are asked: what single change to the model could prevent the electron from radiating energy? They write individual hypotheses. |
| 12 min | The teacher introduces Bohr's postulate: electrons occupy fixed energy levels (stationary states) labelled K, L, M, N, each with a definite energy; in a stationary state no energy is radiated. Students map this to their own hypotheses — whose idea came closest? — and discuss why the K-shell is the lowest energy level. |
| 15 min | Students work in pairs on a structured task: given that shells are labelled K, L, M, N ($n = 1, 2, 3, 4$) with energy increasing outward, and that electrons can only move between shells by absorbing or emitting energy equal to the difference between levels, they draw and annotate three scenarios — electron absorbs energy, electron emits energy, electron stays put — and identify what happens in each case. |
| 13 min | A brief class discussion addresses the K, L, M, N naming convention (Barkla's X-ray spectroscopy origin; lower letters reserved). Students then evaluate: does Bohr's model fully resolve the stability problem, or does it only restate it as a postulate? They write a one-sentence verdict with a reason drawn from the model's content. |

Homework: Draw the shell diagram for a hydrogen atom and for a helium atom; annotate each shell with its label and the number of electrons it holds.

Assessment**Q3 · Learning Outcome****Cognitive Demand**

Students can represent energy transitions between quantised levels using a model and assess whether a postulate resolves the problem it was introduced to solve.

Understanding

Rutherford's model predicted that orbiting electrons should continuously lose energy and spiral into the nucleus. Bohr resolved this by proposing stationary states. Which statement correctly describes what Bohr's stationary state postulate says about an electron in such a state?

- A. The electron can only exist in the K shell, where it is too close to the nucleus to radiate energy effectively.
- B. The electron is stationary — it has stopped moving — so it does not accelerate and therefore does not radiate energy.
- C. **The electron moves in a fixed orbit of definite energy and does not radiate energy while remaining in that orbit. ✓**
- D. The electron radiates a small, constant amount of energy that is replenished by the nucleus, so net energy loss is zero.

Answer section

Correct answer: C

What each choice reveals

- B Confuses 'stationary state' (fixed-energy orbit) with 'stationary electron' (not moving) — the most common linguistic misconception about Bohr's term.
- D Invents a compensating mechanism not present in Bohr's model; reveals a misunderstanding of the postulate as an energy-balance argument rather than a non-radiation assertion.
- A Incorrectly restricts the stationary-state condition to the K shell only; conflates the concept of lowest energy with the only allowed state.

SECTION 4

8.3 What Components Contribute to the Mass of an Atom? / 8.4 Symbols of Elements / 8.6 Mass Number

Period 4

50 min

Protons, Neutrons, Atomic Symbols, and the Mass Number Decoded

Materials: Textbook (subatomic particle table, element symbol table), Blackboard, Index cards (one per student)

Teacher notes: Having established Bohr's shell model, this unit consolidates the full subatomic inventory — protons, neutrons, electrons — and attaches numbers to them through atomic number, mass number, and standard notation. A frequent error is conflating mass number with atomic mass (in u); stress that A is a count of nucleons, not a measurement in grams or atomic mass units. The index-card swap is a quick self-check: if several students make the same error reconstructing neutron count, pause and rework the $A - Z$ relationship on the board before closing.

- 12 min The teacher poses the mass puzzle: helium has 2 protons yet weighs roughly 4 times hydrogen (1 proton) — what accounts for the extra mass? Students predict in writing. The teacher then introduces neutrons (Chadwick, 1932), using the particle table to show that neutrons have mass nearly equal to a proton but carry no charge; the mass discrepancy is resolved.
- 12 min The teacher presents the Berzelius / IUPAC symbol system: first letter capitalised, second (if any) lowercase, some derived from Latin names. Students apply the rules to a short mixed set — given element names, write the correct symbol; given symbols, identify which rule each uses. They flag symbols that derive from non-English names and state the source language.
- 14 min Mass number A (protons + neutrons) is introduced with the standard notation placing A as a superscript and Z as a subscript to the left of the symbol. Students work a set of numerical problems: given A and Z, find neutron count; given neutron count and Z, find A; reconstruct the nuclide notation for hydrogen, helium, and lithium. They verify their answers against the textbook table.
- 12 min Students receive an index card and write on one side a nuclide notation of their choice (choosing from elements covered), and on the other side the full particle inventory (proton, neutron, electron counts). Cards are swapped: the receiving student reconstructs all three counts from the notation and checks against the answer side. Errors are discussed as a class.

Homework: For the elements sodium ($Z = 11$, $A = 23$) and chlorine ($Z = 17$, $A = 35$), write the full nuclide notation and list the number of protons, neutrons, and electrons for each.

Assessment

Q4 · Learning Outcome

Cognitive Demand

Students can apply a symbolic notation system to determine the subatomic particle composition of an atom from given nuclear data.

Application

An atom of phosphorus is represented as $^{31}_{15}\text{P}$.

- (a) State the number of protons, neutrons, and electrons in this atom. Show the calculation for neutron count.
- (b) A second atom has the symbol $^{27}_{13}\text{Al}$. Write the full nuclide notation for an atom of the same element that has 14 neutrons instead of 14 — wait, this atom has 14 neutrons. Calculate its mass number and write the corrected nuclide notation.
- (c) Explain in one sentence why the symbol for iron is Fe and not Ir, using the IUPAC rule for symbols derived from non-English names.

Answer section

Expected elements

- Phosphorus: protons = 15 (read directly from Z), electrons = 15 (atom is neutral so equals proton count), neutrons = $A - Z = 31 - 15 = 16$; calculation shown.
- For aluminium with 14 neutrons: $A = Z + \text{neutrons} = 13 + 14 = 27$; nuclide notation $^{27}_{13}\text{Al}$ — same as the given atom, demonstrating correct application of $A = Z + n$.
- Fe derives from the Latin name 'ferrum'; IUPAC allows symbols from Latin, Greek, or German names when the element was named in those languages, so the symbol need not match the English name.

SECTION 5

8.7 How Are Electrons Distributed in Different Energy Levels?

Period 5

50 min

Filling the Shells: Electronic Configuration for the First Eighteen Elements

Materials: Textbook (Fig. 8.11 and shell-filling table), Plain paper for shell diagrams, Pencils

Teacher notes: Having introduced atomic number and mass number, this unit applies them to the practical work of distributing electrons across shells. The most common error is students placing more than 8 electrons in the outermost shell, even when M can theoretically hold 18 — reinforce that the outermost-shell cap of 8 is a separate rule that overrides the $2n^2$ formula for the outermost position. The sodium checkpoint at $Z = 11$ is a reliable diagnostic: a student who writes 2, 9 has misapplied the outermost-shell rule and needs correction before attempting the elements beyond.

- 10 min The teacher states the Bohr-Bury rules: maximum electrons in shell n is $2n^2$ ($K = 2$, $L = 8$, $M = 18$); outermost shell holds no more than 8; electrons fill from innermost outward. Students apply just the formula to generate the three numbers themselves rather than being told them, checking the arithmetic before writing anything down.
- 20 min Working individually, students draw concentric-circle diagrams for each element from hydrogen ($Z = 1$) to argon ($Z = 18$), distributing electrons shell by shell according to the rules. They annotate each diagram with the element symbol and the configuration notation (e.g., 2, 8, 6 for sulfur). A target-check at sodium ($Z = 11$): students verify they get 2, 8, 1 before continuing.
- 12 min Students compare their completed diagrams with a partner, resolve any discrepancies, and then answer three targeted questions: which elements have a completely filled outermost shell (identify neon and argon and state their configurations); which element first fills the L shell completely; and what is the configuration of silicon ($Z = 14$)?
- 8 min The class reconstructs the configuration of three nuclides given in standard notation ($^{12}_6\text{C}$, $^{19}_9\text{F}$, $^{28}_{14}\text{Si}$) by reading Z from the notation, then applying the filling rules. Students identify the number of valence electrons for each — a direct preparation for valency reasoning.

Homework: Write the electronic configuration and draw the shell diagram for phosphorus ($Z = 15$) and calcium ($Z = 20$); label the valence shell and count the valence electrons for each.

Assessment

Q5 · Learning Outcome

Students can apply a rule-based system to construct the electron-shell distribution of an atom and extract the number of valence electrons from the result.

Cognitive Demand

Application

Use the Bohr-Bury shell-filling rules to answer the following.

- (a) Write the electronic configuration of sulfur ($Z = 16$) and silicon ($Z = 14$). Show the distribution across K, L, and M shells.
- (b) State the number of valence electrons for each element.
- (c) A student writes the configuration of chlorine ($Z = 17$) as 2, 8, 7. A second student writes it as 2, 9, 6. Identify which is correct and explain the error in the incorrect configuration.

Answer section

Expected elements

- Sulfur ($Z = 16$): K = 2, L = 8, M = 6; configuration 2, 8, 6; valence electrons = 6.
- Silicon ($Z = 14$): K = 2, L = 8, M = 4; configuration 2, 8, 4; valence electrons = 4.
- Correct configuration for chlorine is 2, 8, 7 (first student). The second student's error is placing 9 electrons in the L shell, which violates the $2n^2$ rule: for $n = 2$, maximum is $2(2^2) = 8$. Once L is full at 8, the remaining electrons go into M.

SECTION 6

8.8 Combining Capacity of an Atom: Valency

Period 6

50 min

Valency from Electron Counts: Who Loses, Gains, or Shares?

Materials: Textbook (Table 8.4), Blackboard, Worksheet with element configurations

Teacher notes: The electronic configurations built in the electron-distribution work feed directly into this unit — valency is read off the valence electron count, so that content is the entry point here. A common misconception is that valency and valence electron count are always the same number; point out carbon (4 valence electrons, valency 4 by sharing) versus chlorine (7 valence electrons, valency 1 by gaining), and ask students to articulate why the count and the valency differ. The novel-element task at the end probes whether students can apply the rules generatively rather than just look them up from a table.

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| 10 min | The teacher poses an opening question: neon (2, 8) and argon (2, 8, 8) are unreactive — sodium (2, 8, 1) reacts readily — what is different about their outermost shells? Students reason from the electron counts generated in the previous unit and articulate the octet rule before it is formally stated. |
| 15 min | The teacher introduces the three pathways to a complete octet: losing electrons (sodium loses 1 → valency 1), gaining electrons (oxygen gains 2 → valency 2), sharing electrons (carbon shares 4 → valency 4). Students apply this to a set of six elements — magnesium, chlorine, nitrogen, fluorine, aluminium, and sulfur — determining valency from configuration and identifying the likely pathway for each. |
| 13 min | Students extend Table 8.4 with a valency column for all eighteen elements from the previous unit, using the losing/gaining/sharing rules. They compare entries: which elements share the same valency and why? They identify the noble gases (valency 0) and explain the basis of that assignment. |
| 12 min | Groups are given three novel elements described only by their electronic configuration (not named); they determine valency and predict whether each would more likely form a compound with sodium or with chlorine, justifying with the electron-count logic. One group presents; the class evaluates the reasoning. |

Assessment

Q6 · Learning Outcome

Students can determine the combining capacity of an atom by reasoning from its electron configuration to the pathway by which it achieves a complete outer shell.

Cognitive Demand

Understanding

Magnesium has the electronic configuration 2, 8, 2. Which statement correctly identifies its valency and the pathway by which it achieves a complete outer shell?

- A. Valency 0, because magnesium's total electron count (12) is an even number, making it stable.
- B. Valency 2, because magnesium loses its 2 outermost electrons, leaving the L shell (8 electrons) as the new outer shell. ✓**
- C. Valency 2, because magnesium shares 2 electrons with another atom, adding 2 to reach an octet in the M shell.
- D. Valency 6, because magnesium needs to gain 6 electrons to complete its outermost shell to 8.

Answer section

Correct answer: B

What each choice reveals

- D** Applies the gaining pathway instead of the losing pathway; the student counts 6 electrons needed to fill M to 8 without recognising that losing 2 is far easier and is the actual pathway for metals with few valence electrons.
 - C** Correctly identifies valency as 2 but assigns the wrong pathway (sharing rather than losing); the student knows the number but not the mechanism.
 - A** Invents a stability rule based on even electron count — a fabricated heuristic with no basis in the octet rule or the section's content.
-